NATIONAL UNIVERSITY OF LESOTHO

BSc, B. Eng. Examinations

CS3541-14 Computer Communication and Networks 1

December 2024 **Marks: 100** **Time: 3 Hours**

Instructions

1. This is a *closed-book* examination.
2. **Question 1 is compulsory.**
3. Choose ***any 4*** other questions from questions 2 to 7.
4. Each question carries 20 marks.
5. Each question **MUST** begin on a new page.

Question 1 (Compulsory)

1.1 You have just been employed as a new network manager, and you have been given the task of ordering new network equipment for a small business with a large growth forecast. Which primary factor should you be concerned with mostly when choosing this equipment?
(a) devices with a fixed number and type of interfaces
(b) redundant devices
(c) devices that have support for network monitoring
(d) devices with support for modularity

1.2 What is the objective of a network reconnaissance attack?
(a) discovery and mapping of systems
(b) denying access to resources by legitimate users
(c) disabling network systems or services
(d) unauthorized manipulation of data

1.3 A user calls to report that a PC cannot access the internet. The network technician asks the user to issue the command **ping 127.0.0.1** in a command prompt window. The user reports that the result is four positive replies. What conclusion can be drawn based on this connectivity test?
(a) the PC can access the network. The problem exists beyond the local network.
(b) the TCP/IP implementation is functional.
(c) the PC can access the Internet. However, the web browser may not work.
(d) the IP address obtained from the DHCP server is correct.

1.4 A small company has only one router as the exit point to its ISP. Which solution could be adopted to maintain connectivity if the router itself, or its connection to the ISP, fails?
(a) add more interfaces to the router that is connected to the internal network.
(b) purchase a second least-cost link from another ISP to connect to this router.
(c) activate another router interface that is connected to the ISP, so the traffic can flow through it.
(d) have a second router that is connected to another ISP.

1.5 Which of the following characteristics describes crosstalk?
(a) distortion of the network signal from fluorescent lighting
(b) weakening of the network signal over long cable lengths
(c) distortion of the transmitted messages from signals carried in adjacent wires
(d) loss of wireless signal over excessive distance from the access point

1.6 When a router receives a packet, what information must be examined in order for the packet to be forwarded to a remote destination?
(a) source MAC address
(b) source IP address
(c) destination MAC address
(d) destination IP address

1.7 Which statement is true about variable-length subnet masking?
(a) bits are returned, rather than borrowed, to create additional subnets.
(b) subnets may only be subnetted one additional time.
(c) each subnet is the same size.

(d) the size of each subnet may be different, depending on requirements.

1.8 Which field in an IPv6 packet is used by the router to determine if a packet has expired and should be dropped?
(a) address unreachable field
(b) hop Limit
(c) packet Length
(d) time-to-Live

1.9 Although CSMA/CD is still a feature of Ethernet, why is it no longer necessary?
(a) it has been replaced by CSMA/CA.
(b) the use of Gigabit Ethernet speeds.
(c) the virtually unlimited availability of IPv6 addresses.
(d) the use of full-duplex capable Layer 2 switches.

1.10 How does a networked server manage requests from multiple clients for different services?
(a) ach request has a combination of source and destination port numbers, coming from a unique IP address.
(b) each request is tracked through the physical address of the client.
(c) the server uses IP addresses to identify different services.
(d) the server sends all requests through a default gateway.

1.11 Source host A sends a large file to the destination host B over a TCP connection. If the sequence number of this connection is *k*, then the sequence number for the subsequent segment will be *k + 1*.
(a) false
(b) true

1.12 Why would a technician enter the command copy startup-config running-config?
(a) To remove all configurations from the switch
(b) To save an active configuration to NVRAM
(c) To copy an existing configuration into RAM
(d) To make a changed configuration the new startup configuration

1.13 A wireless LAN is being deployed inside a new hall at a guest house. The hall is located at a slightly higher part of the guest house premises. After network testing is complete, the network technician reports that the wireless LAN signal is occasionally affected by some type of interference. What is a possible cause of the signal distortion?
(a) the large number of trees that surround the office
(b) the microwave oven
(c) the elevated location where the wireless LAN was installed
(d) the number of wireless devices that are used in the wireless LAN

1.14 Which network design consideration would be more important to a large corporation than to a small business?
(a) firewall
(b) low port density switch
(c) redundancy
(d) internet router

1.15 Network congestion has resulted in the source learning of the loss of TCP segments that were sent to the destination. What is one way that the TCP protocol addresses this?
(a) the destination decreases the window size.
(b) the source decreases the window size to decrease the rate of transmission from the destination.
(c) the source decreases the amount of data it transmits before it receives an acknowledgment from the destination.
(d) the destination sends fewer acknowledgments in order to conserve bandwidth.

1.16 What type of transmission is involved in communication between a computer and a keyboard?
(a) automatic and dual
(b) half-duplex
(c) full-duplex
(d) simplex

1.17 What is the binary representation of 0xCA?
(a) 10111010
(b) 11010101
(c) 11001010
(d) 11011010

1.18 What process takes multiple streams of segmented data and interleaves them together?
(a) encoding
(b) access control
(c) layering
(d) multiplexing

1.19 What type of attack may involve the use of tools such as nslookup and fping**?**
(a) denial of service attack
(b) reconnaissance attack
(c) worm attack
(d) access attack

1.20 Which statement is true about CDP on a Cisco device?
(a) Because it runs at the data link layer, the CDP protocol can only be implemented in switches.
(b) To disable CDP globally, the no cdp enable command in interface configuration mode must be used.
(c) CDP can be disabled globally or on a specific interface.
(d) The show CDP neighbor detail command will reveal the IP address of a neighbor only if there is Layer 3 connectivity.

[20 marks]

CHOOSE FOUR QUESTIONS FROM QUESTION 2 TO QUESTION 7

Question 2

(a) Show the layers of the TCP/IP protocol stack that are implemented on the following network components, in the correct order: **(3)**
   (i) End device
   (ii) Router
   (iii) Switch

(b) Briefly explain what ICMP is and give two examples of its key areas of functionality. **(3)**

(c) Write down three features that are present in ICMPv6 but not in ICMPv4 and explain the role of each feature. **(6)**

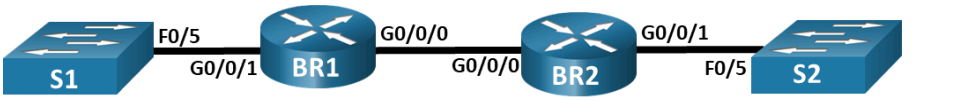
(d) Explain how the two IPv4 to native IPv6 migration techniques can be implemented. You can use diagrams, **(8)**

[20 marks]

Question 3

(a) Mpopo uses his web-based electronic mail account to send a message to his friend Moiloa, who accesses her electronic mail from a mail server using IMAP. Outline a series of application layer protocols involved to move the messages between the two hosts. **(3)**

(b) Apply variable-length subnetting to segment the **192.168.33.128/25** network to fulfill the IPv4 addressing requirements of the given topology.



BR1 LAN: 40 Hosts

BR1-BR2 Link: 2 Hosts

BR2 LAN: 25 Hosts
BR2 IoT LAN (Future): 5 Hosts
BR2 CCTV LAN (Future): 4 Hosts
BR2 HVAC C2LAN (Future): 4 Hosts

Show how you arrive at your answers and then present the answers using the table below. **(9)**

Subnet Name	Maximum number of hosts	Network address	First host address	Last host address	Broadcast address

(c) Using the common IP addressing scheme discussed in class, assign IPv4 addresses to all router interfaces. Present your answer in a table similar to the one below: **(8)**

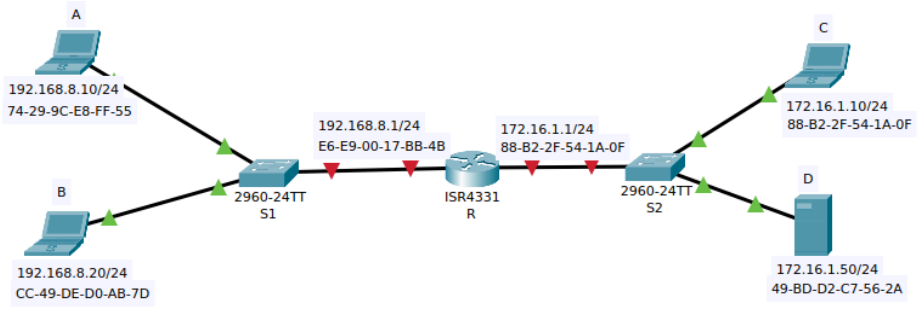
Device	Interface	IP Address	Subnet Mask
BR1			
BR2	G0/0/0		

	G0/0/1		

[20 marks]

Question 4

Consider the topology shown in the figure below.



(a) Briefly explain how S1 switches packets. **(2)**

(b) Explain the role of ARP for the following scenarios:

(i) Between A and B **(2)**

(ii) Between B and D. **(1)**

(c) Write down the layer 2 and layer 3 source and destination addresses when the submitted request is at the following points on its journey to its destination:

(i) A **(2)**

(ii) R **(2)**

(d) Briefly explain how the domain name service is implemented and how DNS queries are resolved. **(4)**

(e) Explain the multiple access problem in computer networks. **(2)**

(f) A local bank intends to connect one of its branches located in Berea district to the main office network in Maseru. The service provider offers the following options: Dedicated Leased Line, ADSL, and Satellite.

(i) Choose the best option that the bank can use and motivate your choice. **(3)**

(ii) For technologies not chosen, explain why they are deemed inappropriate. **(2)**

[20 marks]

Question 5

(a) Suppose the source sends two TCP segments back to back to the destination host over a TCP connection. The first segment has a sequence number 440, the second has a sequence number 500.

(i) Calculate the payload (in bytes) in the first segment. **(2)**

(ii) Under what circumstance (s) will the destination send a duplicate ACK number? **(2)**

(b) Differentiate between unicast and broadcast delivery options. Also, give an example frame destination address for each option. **(2)**

(c) Compare HTTP and SMTP application layer protocols along the following dimensions:

(i) Encapsulation **(4)**
(ii) Push vs pull **(4)**
(iii) Status codes **(2)**
(iv) Message headers **(2)**
(v) Transport protocol **(2)**

[20 marks]

Question 6

(a) Differentiate between global unicast address and link-local address in IPv6. **(4)**
(b) Explain the dynamic process of generating IPv6 link-local addresses. **(3)**
(c) Given the 2001:db8:acad::/48 global routing prefix with a 16-bit subnet ID, write down the first eleven subnets of this network. **(5)**
(d) Explain how TCP manages flow control. **(4)**
(e) Network security should always be a priority in modern-day computer networks.
(i) Differentiate between a trojan horse and a virus attack. (**2)**
(ii) Using examples, differentiate between internal and external network attacks. **(2)**

[20 marks]

Question 7

(a) TCP uses a connection-oriented mechanism to achieve its functions. Explain what this mechanism is called and what it achieves. **(3)**
(b) With the aid of a diagram, explain how the mechanism in (a) works. **(6)**
(c) The network layer in the source host accepts a segment of 1,200 bytes and a destination host address from the transport layer. Suppose many network application processes run at the destination host at the same time.
(i) Describe an appropriate transport-layer protocol that will get application data to the desired process at the destination, assuming the operating system in the destination host has assigned a 4-byte port number to each running application process. **(6)**
(ii) Modify the protocol in (i) so that it provides a return address to the destination process. **(5)**

[20 marks]